

## Docker Commands Cheat Sheet with Examples



### Basic Docker Commands

#### 1. Check Docker Version

- Command: `docker --version`
- Example: `docker --version` → Outputs: Docker version 20.10.7

#### 2. Run a Container

- Command: `docker run <image-name>`
- Example: `docker run nginx` → Runs an NGINX container.

#### 3. List Running Containers

- Command: `docker ps`
- Example: `docker ps` → Shows all running containers.

#### 4. List All Containers

- Command: `docker ps -a`
- Example: `docker ps -a` → Shows running and stopped containers.

#### 5. Stop a Container

- Command: `docker stop <container-id>`
- Example: `docker stop 12345abc` → Stops the specified container.

#### 6. Remove a Container

- Command: `docker rm <container-id>`
- Example: `docker rm 12345abc` → Removes the specified container.

## 7. Pull an Image

- Command: `docker pull <image-name>`
- Example: `docker pull ubuntu` → Downloads the Ubuntu image.

## 8. List Docker Images

- Command: `docker images`
- Example: `docker images` → Lists all Docker images on the host.

## 9. Remove an Image

- Command: `docker rmi <image-id>`
- Example: `docker rmi abc123` → Removes the specified image.

## 10. Build an Image from a Dockerfile

- Command: `docker build -t <image-name> .`
- Example: `docker build -t my-app .` → Builds an image named "my-app."

## 11. View Logs of a Container

- Command: `docker logs <container-id>`
- Example: `docker logs 12345abc` → Shows logs of the specified container.

## 12. Run a Command in a Running Container

- Command: `docker exec <container-id> <command>`
- Example: `docker exec 12345abc ls /app` → Lists files in /app.

## 13. Start a Stopped Container

- Command: `docker start <container-id>`
- Example: `docker start 12345abc` → Restarts a stopped container.

## 14. Restart a Container

- Command: `docker restart <container-id>`
- Example: `docker restart 12345abc` → Restarts the specified container.

## 15. Inspect a Container

- Command: `docker inspect <container-id>`
- Example: `docker inspect 12345abc` → Displays detailed information about the container.

## 16. Inspect an Image

- Command: `docker inspect <image-name>`
- Example: `docker inspect ubuntu` → Shows details about the Ubuntu image.

### 17. Export a Container to a Tar File

- Command: `docker export <container-id> > <file-name>.tar`
- Example: `docker export 12345abc > container.tar` → Exports container filesystem.

### 18. Import a Tar File as an Image

- Command: `docker import <file-name>.tar`
- Example: `docker import container.tar` → Creates an image from the tar file.

### 19. Run a Container in Detached Mode

- Command: `docker run -d <image-name>`
- Example: `docker run -d nginx` → Runs NGINX in the background.

### 20. Run a Container with Port Mapping

- Command: `docker run -p <host-port>:<container-port> <image-name>`
- Example: `docker run -p 8080:80 nginx` → Maps host port 8080 to container port 80.

### 21. Check Docker System Information

- Command: `docker info`
- Example: `docker info` → Displays Docker environment details.

### 22. Save an Image as a Tar File

- Command: `docker save -o <file-name>.tar <image-name>`
- Example: `docker save -o my-app.tar my-app` → Saves "my-app" image as a tar file.

### 23. Load an Image from a Tar File

- Command: `docker load < <file-name>.tar`
- Example: `docker load < my-app.tar` → Loads "my-app" image from the tar file.

### 24. Tag an Image

- Command: `docker tag <image-id> <repository>:<tag>`
- Example: `docker tag abc123 my-app:latest` → Tags image as "my-app:latest."

### 25. Push an Image to Docker Hub

- Command: `docker push <repository>:<tag>`
- Example: `docker push my-app:latest` → Pushes "my-app:latest" to Docker Hub.

### 26. Create a Volume

- Command: `docker volume create <volume-name>`
- Example: `docker volume create my-volume` → Creates a volume named "my-volume."

### 27. List Volumes

- Command: `docker volume ls`
- Example: `docker volume ls` → Lists all volumes.

### 28. Remove a Volume

- Command: `docker volume rm <volume-name>`
- Example: `docker volume rm my-volume` → Removes "my-volume."

### 29. Create a Network

- Command: `docker network create <network-name>`
- Example: `docker network create my-network` → Creates a network named "my-network."

### 30. List Networks

- Command: `docker network ls`
- Example: `docker network ls` → Lists all Docker networks.

### 31. Connect a Container to a Network

- Command: `docker network connect <network-name> <container-id>`
- Example: `docker network connect my-network 12345abc` → Connects container to "my-network."

### 32. Disconnect a Container from a Network

- Command: `docker network disconnect <network-name> <container-id>`
- Example: `docker network disconnect my-network 12345abc` → Disconnects container from "my-network."

### 33. Remove a Network

- Command: `docker network rm <network-name>`
- Example: `docker network rm my-network` → Removes "my-network."

### 34. Create a Detached Container with a Name

- Command: `docker run -d --name <container-name> <image-name>`
- Example: `docker run -d --name my-nginx nginx` → Runs NGINX container named "my-nginx."

### 35. Rename a Container

- Command: `docker rename <old-name> <new-name>`
- Example: `docker rename old-container new-container` → Renames a container.

### 36. Prune Unused Docker Objects

- Command: `docker system prune`
- Example: `docker system prune` → Cleans up unused containers, images, and volumes.

### 37. Inspect a Volume

- Command: `docker volume inspect <volume-name>`
- Example: `docker volume inspect my-volume` → Displays details about "my-volume."

### 38. Inspect a Network

- Command: `docker network inspect <network-name>`
- Example: `docker network inspect my-network` → Shows details of "my-network."

### 39. Limit CPU Usage for a Container

- Command: `docker run --cpus="<value>" <image-name>`
- Example: `docker run --cpus="1.5" nginx` → Limits the container to 1.5 CPUs.

### 40. Limit Memory Usage for a Container

- Command: `docker run -m <value> <image-name>`
- Example: `docker run -m 512m nginx` → Limits memory to 512 MB.

### 41. Run an Interactive Container

- Command: `docker run -it <image-name>`
- Example: `docker run -it ubuntu bash` → Starts an interactive Ubuntu shell.

### 42. Kill a Running Container

- Command: `docker kill <container-id>`
- Example: `docker kill 12345abc` → Forcefully stops the container.

### 43. Create a Dockerfile

- Command: `touch Dockerfile`
- Example: `touch Dockerfile` → Creates a new Dockerfile in the current directory.

### 44. Login to Docker Hub

- Command: `docker login`
- Example: `docker login` → Prompts for Docker Hub credentials.

### 45. Logout from Docker Hub

- Command: `docker logout`
- Example: `docker logout` → Logs out from Docker Hub.

### 46. Run a Container with Environment Variables

- Command: `docker run -e <env-var>=<value> <image-name>`
- Example: `docker run -e ENV=production nginx` → Sets "ENV" variable to "production."

#### 47. View Container Stats

- Command: `docker stats`
- Example: `docker stats` → Displays resource usage statistics of running containers.

#### 48. Change Restart Policy of a Container

- Command: `docker update --restart=<policy> <container-id>`
- Example: `docker update --restart=always 12345abc` → Updates the restart policy.

#### 49. Pause a Running Container

- Command: `docker pause <container-id>`
- Example: `docker pause 12345abc` → Pauses the container.

#### 50. Unpause a Paused Container

- Command: `docker unpause <container-id>`
- Example: `docker unpause 12345abc` → Resumes the paused container.

#### 51. List All Swarm Nodes

- Command: `docker node ls`
- Example: `docker node ls` → Lists all nodes in a Docker Swarm cluster.

#### 52. Create a Service in Swarm Mode

- Command: `docker service create --name <service-name> <image-name>`
- Example: `docker service create --name web nginx` → Creates a service named "web" running NGINX.

#### 53. List Services in Swarm Mode

- Command: `docker service ls`
- Example: `docker service ls` → Lists all services in a Swarm cluster.

#### 54. Inspect a Docker Service

- Command: `docker service inspect <service-name>`
- Example: `docker service inspect web` → Displays details of the "web" service.

#### 55. Scale a Service

- Command: `docker service scale <service-name>=<replicas>`
- Example: `docker service scale web=5` → Scales the "web" service to 5 replicas.

#### 56. Update a Service

- Command: `docker service update --image <new-image> <service-name>`
- Example: `docker service update --image nginx:latest web` → Updates the "web" service with a new image.

#### 57. Remove a Service

- Command: `docker service rm <service-name>`

- Example: `docker service rm web` → Removes the "web" service.

#### 58. Initialize Docker Swarm

- Command: `docker swarm init`
- Example: `docker swarm init` → Initializes a Swarm cluster.

#### 59. Join a Node to a Swarm

- Command: `docker swarm join --token <token> <manager-IP>`
- Example: `docker swarm join --token abc123 192.168.1.10:2377` → Joins a node to a Swarm cluster.

#### 60. Leave a Swarm Cluster

- Command: `docker swarm leave`
- Example: `docker swarm leave` → Removes a node from the Swarm cluster.

#### 61. Promote a Node to Manager

- Command: `docker node promote <node-id>`
- Example: `docker node promote worker1` → Promotes "worker1" to a manager.

#### 62. Demote a Manager Node

- Command: `docker node demote <node-id>`
- Example: `docker node demote manager1` → Demotes "manager1" to a worker.

#### 63. List Tasks in Swarm Mode

- Command: `docker service ps <service-name>`
- Example: `docker service ps web` → Lists tasks associated with the "web" service.

#### 64. Display Events from Docker Daemon

- Command: `docker events`
- Example: `docker events` → Displays real-time events from Docker daemon.

#### 65. Display Disk Usage by Docker

- Command: `docker system df`
- Example: `docker system df` → Shows disk usage by images, containers, and volumes.

#### 66. Enable BuildKit

- Command: `DOCKER_BUILDKIT=1 docker build -t <image-name> .`
- Example: `DOCKER_BUILDKIT=1 docker build -t my-app .` → Builds image using BuildKit.

#### 67. Create a Context

- Command: `docker context create <context-name>`
- Example: `docker context create my-context` → Creates a new Docker context.

#### 68. List Docker Contexts

- Command: `docker context ls`
- Example: `docker context ls` → Lists all Docker contexts.

#### 69. Use a Docker Context

- Command: `docker context use <context-name>`
- Example: `docker context use my-context` → Switches to "my-context."

#### 70. Remove a Docker Context

- Command: `docker context rm <context-name>`
- Example: `docker context rm my-context` → Removes "my-context."